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MINISTRY OF HEALTH OF THE RUSSIAN FEDERATION Federal
State Budgetary Institution

**'DMITRY ROGACHEV NATIONAL MEDICAL RESEARCH CENTRE FOR
PAEDIATRIC HAEMATOLOGY, ONCOLOGY AND IMMUNOLOGY'**

of the Ministry of Health of the Russian Federation
(Federal State Budgetary Institution 'Dmitry Rogachev National Medical
Research Centre for Paediatric Haematology, Oncology and Immunology' of the
Ministry of Health of the Russian Federation)

3 April 2026

№ 1480

Moscow

**EXTRACT
FROM MEDICAL RECORD No. 2026/4452**

Patient's full name: **EGOR ANDREEVICH BARAKHTEIKO**

Date of birth: **24 April 2008**

Age: 18 years

Gender: **Male**

Registered address: Russian Federation, Sakha (Yakutia) Republic, Yakutsk, III
Sergelyakhskoye, 5 km, house 45, flat 5A

Current address: Russian Federation, Sakha (Yakutia) Republic, Yakutsk, 5 km along III
Sergelyakhskoye Road, house 45, flat 5A

Admitted to Haematopoietic Stem Cell Transplantation Unit No. 1 from 24 April 2026 to 3
May 2026

Clinical diagnosis:

Primary: Myelodysplastic syndrome with WT1 overexpression, NRAS gene mutation from 21 November 24; first allogeneic haematopoietic stem cell transplant (HSCT) from an unrelated HLA-identical donor on 1 April 25; first clinical-haematological and molecular remission from 29 April 25. Secondary myeloid leukaemia following myelodysplastic syndrome from 3 July 25, an internal tandem duplication in the FLT3 gene (FLT3/ITD), WT1 gene overexpression, BAALC; repeat allogeneic HSCT from a haploidentical donor (father) on 16 September 25; clinical-haematological and molecular remission from 14 October 25; molecular relapse (FLT3/ITD, overexpression of the WT1 gene, BAALC-39) from 13 November 25, persistence of minimal residual disease (MRD) (IFT) from 4 December 25; infusion of donor lymphocytes No. 1 (1.0×10^5) on 8 December 25; bone marrow relapse from 15 January 26. Administration of anti-CD33 CAR-NK lymphocytes on 8 April 2026 from a haploidentical donor (father). Progression on 23 April 2026. (C92.0 Acute myeloid leukaemia [AML])

Complication of the underlying condition: Recipient of TGSK (D84.8 Other specified immunodeficiency disorders)

Complication of the underlying condition: Recipient of TGSK (Z94.8 Presence of other transplanted organs and tissues)

Complication of the main condition: Transplant-induced haematopoietic aplasia (D70 Agranulocytosis)

Concomitant condition: Left aorto-mesenteric conflict (compression of the left renal vein). (N28.8 Other specified diseases of the kidney and ureter)

Blood group **0(I)Rh+** Date of determination **25 March 2026**

Life history: Child from the first pregnancy and first delivery. Birth weight 3,600 g, height 53 cm. Discharged from the maternity ward to go home. Vaccinated according to the national immunisation schedule up to the age of 16; not vaccinated after the TGSK. Past illnesses: chickenpox in 2013, umbilical hernia in 2016 (operated on), acute respiratory viral infections 2–3 times a year. Ingrown toenails on the big toes of both feet.

Family history: Sibling, a girl born in 2010 – healthy. Full sister born in 2024 – healthy. No family history of haematological or oncological diseases.

Allergy history: lactase deficiency?

Surgeries and injuries: In 2013, he underwent surgery for hydrocele of the testicles
2016: herniorrhaphy – umbilical hernia, circumcision.
29 July 2025: necrectomy for paronychia.
13 March 26: bilateral maxillary sinus puncture

Blood transfusion history: Blood component transfusions were carried out without complications

Medical history: the patient has been unwell since September 2024, when he began complaining of weakness and rapid fatigue; the following investigations were carried out:

- Complete blood count dated 20 November 2024: white blood cells 12.9 thousand/ μ l, haemoglobin 58 g/l, platelets 37 thousand/ μ l; blasts 3 per cent.
 - bone marrow aspiration dated 21 November 2024: the bone marrow aspiration revealed a hypercellular bone marrow with blast cells ranging from 11.4% to 13.8%. Erythropoiesis is predominantly of the normoblastic type with a small megaloblastic component, showing signs of dyserythropoiesis. The granulocytic series is narrowed, with signs of dysgranulopoiesis. Megakaryocytes are present in small numbers without apparent functional activity, with signs of dysmegakaryopoiesis. The morphological picture of the bone marrow is consistent with MDS, IB variant II.
 - Bone marrow immunofluorescence staining dated 21 November 24: a population of cells with low and intermediate granularity was identified, showing reduced expression of the pan-leukocyte marker CD45 and expression of CD34 (11% blasts).
 - Bone marrow trephine biopsy dated 4 December 24: Morphological picture of myeloid neoplasia with marked expansion, rejuvenation and impaired maturation of the granulocytic lineage, with an excess of CD34+ cells and signs of dysplasia in the megakaryocytic lineage.
 - Molecular analysis of bone marrow revealed a mutation in the NRAS gene (somatic).
 - According to DNA sequencing data (dated 17 February 2025): a nucleotide sequence variant was identified in exon 7/10 of the WT1 gene in a heterozygous state
- Thus, based on the findings of the examination, the patient is diagnosed with myelodysplastic syndrome.

• On 10 December 24, hypomethylating therapy (azacitidine) was initiated in combination with venetoclax; following two courses of treatment, clinical and haematological remission was achieved from 15 January 25, and molecular remission from 4 March 25.

• On 1 April 2025, allogeneic HSCT was performed using cells from an unrelated HLA-identical donor (PSKK).

- Conditioning regimen: busulfan 12 mg/kg, fludarabine 150 mg/m².
- Graft-versus-host disease (GVHD) prophylaxis: cyclophosphamide on days +3 and +4, tacrolimus and mycophenolate mofetil from day +5.
- Graft engraftment on day 23.
- Complications in the early post-transplant period: enteropathy of mixed aetiology, acute haemorrhagic cystitis, grade 2 oral mucositis, neurotoxic reaction, toxic dermatitis, acute periodontitis.

According to the results of the bone marrow examination on day +30, the patient remains in complete clinical, haematological and molecular remission of the underlying disease.

On day +92 from TGSK (03.07.25): blast count up to 4 per cent; myelogram (blasts 23.2 per cent); IFT method (blasts 11.4 per cent); PCR method (overexpression of BAALC and WT1 genes).

Thus, based on the examination findings, the patient was diagnosed with transformation into

acute myeloid leukaemia in the early post-transplant period.

From 09.07.25 to 11.07.25, a course of FLA was administered in combination with venclaxab from 12.07.25. Complications:

- sepsis, septic shock.
- soft tissue infection of the first toe of the left foot (*Klebsiella pneumoniae*); necrectomy of tissue from the first toe of the right foot was performed.
- focal pneumonia in segment S6 of the right lung (+ galactomannan)

On day +118 from the start of TGSK (29 July 25):

- bone marrow smear: 70.6% blasts.
- Molecular biology analysis: internal tandem duplication of FLT3 (FLT3/ITD) detected; overexpression of the WT1 and BAALC genes.

On 5 August 25, a lumbar puncture was performed with intrathecal administration of methotrexate and prednisolone; the cerebrospinal fluid was unremarkable.

Given the marked blastosis of the bone marrow, the patient's condition following septic shock, and the presence of the FLT3/ITD mutation in the bone marrow, triple-agent therapy (5-azacitidine, venetoclax, gilteritinib) was initiated.

- Myelogram (04/08/25): the blast count was not increased.

Given the diagnosis and the development of disease relapse following allogeneic haematopoietic stem cell transplantation (HSCT) from an unrelated donor, a decision was made to perform a repeat allogeneic HSCT from a haploidentical donor (the father) with the aim of achieving remission.

On 16 September 25, a repeat allogeneic HSCT from a haploidentical donor (the father) was performed, using PSCC as the graft source.

- Conditioning regimen: melphalan 140 mg/m², fludarabine 150 mg/m².
- Graft-versus-host disease (GVHD) prophylaxis: cyclophosphamide, sirolimus, ruxolitinib.
- Graft engraftment confirmed on day 24.
- Complications in the early post-transplant period: febrile neutropenia, right-sided upper lobe pneumonia (*Enterococcus faecium*, EBV), haemorrhagic cystitis, chronic sphenoid sinusitis

According to the results of the bone marrow examination on day +30 (14 October 2025), complete remission was achieved (the myelogram showed no increase in the number of blasts; no expression of the BAALC gene, no FLT3 internal tandem duplication (FLT3/ITD), no overexpression of the WT1 gene, and no overexpression of the EVI1 gene were detected; cerebrospinal fluid was normal), with complete donor chimerism.

On day +58 after HSCT (13 November 25): molecular relapse of the primary disease was documented (FLT3/ITD, WT1 gene overexpression). Chimerism >97%.

To achieve remission of the primary disease, treatment with gilteritinib was administered from 20 November 25 to 4 December 25.

On day +79 from TGSK (4 December 25):

- bone marrow smear: hypoplastic bone marrow, no increase in the number of blasts.
- BM IFT: persistence of MOB (5%), CD117+ expression.
- Chimerism: decreased to 80–89 per cent.
- Molecular genetic analysis: an internal tandem duplication of FLT3 (FLT3/ITD) was detected, along with overexpression of the WT1 gene.

Given the patient's medical history, the development of molecular relapse, and the persistence of minimal residual disease (MRD; IFT) in the early stages following repeat allogeneic haematopoietic stem cell transplantation (HSCT), the prognosis is poor. A decision was made to initiate combination therapy with FLT3 inhibitors (gilteritinib), BCL2 inhibitors (venetoclax), hypomethylating agents (azacitidine) and a second-generation tyrosine kinase inhibitor (dasatinib).

On 08/12/25, the first infusion of donor lymphocytes (1.0×10^5) was administered.

- From 09.12.25 to 15.12.25, a course of hypomethylating therapy (azacitidine).
 - From 15 December 2025 to 22 December 2025, a course of gilteritinib therapy (80 mg/day).
 - From 29 December 25 to 10 January 26: a course of venetoclax therapy (100 mg/day) (discontinued due to the development of haematological toxicity).
- From 26 December 25 to 15 January 25: a course of treatment with dasatinib (50 mg/day)

On day +121 from TGSK (15 January 26):

- complete blood count: blast count 14%
- bone marrow examination: blasts 27.8%, hypoplastic bone marrow.
- chimerism: reduced to 50–59 per cent
- Molecular genetic analysis: overexpression of the WT1 gene, BAALC-39, and an internal tandem duplication of FLT3 (FLT3/ITD).
- cerebrospinal fluid analysis: no blast cells.
- Thus, a bone marrow relapse of the primary disease was confirmed.

Given the progression of the primary disease despite ongoing comprehensive therapy and the absence of alternative standard treatment methods, a multidisciplinary consultation was held at the D. Rogachev National Medical Research Centre, and a decision was made to proceed with NK-cell therapy; on 16 February 2026, leukapheresis was performed on the father.

Therapy was administered for palliative purposes as follows:

- venetoclax (400 mg) from 15 January 2026, reduced to 200 mg from 2 February 2026 to 12 February 2026.
- cytarabine (50 mg/day) from 24 January 2026, 100 mg/m² (daily infusion) from 29 January 2026, with the dose increased to 160 mg/m² from 3 February 2026 to 27 February 2026.
- dasatinib (70 mg/day) from 29 January 2026 to 13 February 2026.
- azacitidine (75 mg/m²) from 21 January 2026 to 28 January 2026.
- Daunorubicin IV on 30 January 2026.

On day +156 from TGSK (19 February 26):

- bone marrow smear: blasts 18.4% (showing a decrease over time).
- Chimerism: increased to 90–97 per cent.
- Molecular genetic analysis: downregulation of the WT1 gene, upregulation of the BAALC gene, and downregulation of the FLT3 internal tandem duplication (FLT3/ITD).

Complications during the ‘maintenance’ phase of treatment:

1. Grade 1 gastrointestinal mucositis.
2. Bilateral pneumonia (Asp. Ag., Staphylococcus aureus, Klebsiella pneumoniae, CMV, EBV 15 January 2026; EBV, HIV-6, RSV, rhinovirus, Candida, Staphylococcus haemolyticus, Leuconostoc lactis, AsAg6 27 February 2026).
 - Chest CT scan dated 15 January 2026: CT findings showing focal infiltrative changes in both lungs.
 - Chest CT scan dated 14 February 2026: positive progression, with regression of bilateral focal infiltrative changes; a cavity has formed in the lesion in the upper lobe of the right lung.
 - Chest CT scan dated 25 February 2026: adverse progression in the form of new focal changes consistent with bronchiolitis in the lower lobe of the left lung.
 - Chest CT scan dated 9 March 2026: unfavourable progression in the form of bilateral polysegmental areas of pulmonary consolidation and peribronchial dissemination. CT findings consistent with bilateral hydrothorax, hydropericardium and fluid in the abdominal cavity.
 - CT scan of the chest dated 23 March 26: moderate improvement in bilateral focal-infiltrative changes in the lung tissue.
3. Polysinusitis (Enterococcus faecalis, S – vancomycin, linezolid, tigecycline according to the results of the middle ear puncture on 13 March 2026).
4. Bilateral otitis media, with a perforated eardrum on the right.

Combined antimicrobial therapy was administered (ceftazidime + avibactam/meropenem, aztreonam, colistin, tigecycline, levofloxacin, colistin, metronidazole, amphotericin B/voriconazole).

- From 10 March 2026 to 12 March 2026, cytarabine therapy was resumed at a dose of 75 mg/m²/day.

From 11 March 2026, treatment with gilteritinib 120 mg/day has been administered; from 13 March 2026, treatment with venetoclax 200

mg/day.

- On 25 March 2026, the patient was admitted to the isolation ward of the Haematology and Paediatric Oncology Department for an initial routine assessment prior to CAR-T cell therapy and to continue specific treatment.

On admission, his condition was severe due to the underlying disease and complications, including bilateral polysegmental pneumonia (of mixed aetiology), bilateral suppurative otitis media, pansinusitis (*Ent. faecium*) according to the discharge summary from her place of residence; as a result, antimicrobial therapy was initiated comprising meropenem, colistimethate, selezolid and ambizome intravenously. Assessment of the underlying condition:

- Complete blood count dated 25 March 26: white blood cells 0.72 thousand/ μ l, haemoglobin 99 g/l, platelets 81 thousand/ μ l, neutrophils 0.01 thousand/ μ l.
- Differential white blood cell count dated 25 March 2026: absolute neutrophil count 3 per cent, lymphocytes 91 per cent, monocytes 5 per cent, myelocytes 1 per cent
- Myelogram dated 26 March 2026: blast cells I and 10%; all bone marrow haematopoietic lineages are reduced.
- Bone marrow immunofluorescence staining dated 26 March 2026: tumour population 7.5%.
- Cerebrospinal fluid cytology dated 26 March 2026: cytolysis, no blast cells.
- Flow cytometry dated 1 April 26: FLT3-ITD, 85% (ratio = 0.06).

Thus, relapse of the primary disease persists, with a reduction in blast percentage in the bone marrow (previously 18.4% on 19 February 26). As part of 'maintenance' therapy, the patient continues to receive venclesta 200 mg/day, as well as the FLT3 inhibitor gilteritinib (Xospata) 120 mg/day orally. Given the refractory course of secondary acute myeloid leukaemia against the background of previously administered comprehensive specific therapy (including repeat allogeneic haematopoietic stem cell transplantation), and the absence of alternative standard treatment methods, the patient is scheduled to undergo anti-CD33 CAR-NK therapy.

On 30 March 2026, a Broviack-type dual-lumen central venous catheter was successfully inserted without technical difficulties to ensure long-term venous access.

Complications during the observation period: I.

Infectious:

1) Bilateral polysegmental pneumonia (RS virus, parainfluenza) without signs of respiratory failure: combined antimicrobial therapy was administered (meropenem, colistimethate, selezolid, ambizome IV). Bronchoscopy with bronchoalveolar lavage (BAL) on 27 March 26 revealed Parainfluenzae virus III and RSV; no data were obtained regarding the course of fungal or bacterial infection. Given the absence of evidence of Gram-negative infection and persistent afebrile status, colistimethate was discontinued on 31 March 26;

- on 1 April 26, Meronem was replaced with Sulcef; taking into account the absence of evidence of a fungal infection, as well as a rise in creatinine levels (peaking at 176 μ mol/l), Ambizom was replaced with Vifend IV.

2) Bilateral purulent otitis media, right-sided perforated otitis media: combined antimicrobial and symptomatic treatment was administered. According to microbiological analysis of discharge from the right ear, no microorganisms were detected.

3) Pansinusitis (*Ent. faecium* according to the results of the middle ear puncture on 13 March 26): combined antimicrobial and symptomatic treatment was administered. II.

Non-infectious: 1) Acute kidney injury – isolated rise in creatinine (peaking at 176 μ mol/l, eGFR based on cystatin C 68 ml/min/m²): infusion therapy was increased, antimicrobial therapy was de-escalated (Ambizom was discontinued), and ongoing monitoring was continued.

2) Haematopoietic aplasia, agranulocytosis: replacement blood transfusion therapy was administered as required.

According to the examination findings, no absolute contraindications to anti-CD33 CAR-NK therapy were identified.

- On 3 April 2026, the patient was admitted to the TGSK1 ward to undergo CAR-NK cell therapy from a haploidentical donor (his father).

- From 4 April 2026 to 6 April 2026, lymphodensitising therapy was administered:

- Fludarabine 90 mg/m² (SD 160 mg) from day -4 to day -2
- Cyclophosphamide 750 mg/m² (SD 1342 mg) on day -2 + urometixam (SD 2412 mg)

On 8 April 2026, anti-CD33 CAR-NK cells from a haploidentical donor (his father) were administered;

Assessment of remission status:

15 April 2026 (+7 days):

- Bone marrow smear: 30.4/40% blast cells

- MRD (IFT): 22.14% tumour cells
- Persistence of anti-CD33 CAR-NK lymphocytes in blood/BM: 63.9% of 0.1 thousand/ml (NK) / 75.1% of 0.8 thousand/ml (NK)

23 April 2026 (+14 days):

- Myelogram: 71/60.0% blast cells by count, respectively.
- MRD (IFT): tumour population 51.2%; CD33⁺ expression in 100% of blast cells
- Persistence of anti-CD33 CAR-NK lymphocytes: in bone marrow, 35.2 per cent of 0.1 thousand/ml NK cells; in peripheral blood, 40.4 per cent of NK cells;

Thus, on day +14 following the administration of anti-CD33 CAR-NK lymphocytes, the patient's underlying disease has progressed.

Complications during the observation period:

- Bilateral suppurative otitis media, right-sided perforated otitis media, chronic rhinosinusitis – combined antibiotic therapy was administered; the patient was monitored by an ENT specialist; the condition improved over time, and resolution of the infectious foci was confirmed.
- Course of bilateral polysegmental pneumonia (RS virus, parainfluenza) with a positive clinical course, as evidenced by an improvement in the auscultatory findings and radiological signs of organisation of the lesions (MSCT 15 April 2026).
- Acute kidney injury – positive progress in the form of normalisation of plasma creatinine and urea levels, and restoration of glomerular filtration rate.

Virological monitoring of blood samples remained negative

Catheters on admission: Central (non-tunnelled) – 8 March 2026

Physical status on admission: Height: 186.5, Weight: 59.6, Body mass index: 1.76, Heart rate: 80, Respiratory rate: 16, SpO₂: 98, Systolic blood pressure: 115, Diastolic blood pressure: 78, Temperature: 36.4

Complaints: hearing loss (right tympanic membrane perforation on 16 March 2026), cough with sputum.

Condition: severe due to the underlying condition, as well as bilateral focal-infiltrative pneumonia of mixed aetiology (as per chest X-ray findings), pansinusitis and bilateral purulent otitis media.

General well-being: unaffected.

Consciousness: clear.

Neurological status: unremarkable.

Position: active.

Endocrine system: no visible abnormalities.

Body build: asthenic.

Skin: pale pink, clean, dry. No signs of haemorrhagic syndrome. Tissue turgor is adequate. Subcutaneous fat is poorly developed, evenly distributed; no visible oedema.

Mucous membranes: Tongue clean, moist. Mucous membranes clean, pale pink, moist.

Pharyngeal walls not hyperaemic.

Musculoskeletal system: no pathological changes; no visible bone deformities.

Lymphatic system: no peripheral lymphadenopathy.

Cardiovascular system: on auscultation, heart sounds are clear and rhythmic; no pathological murmurs are detected. Blood pressure is normal.

Respiratory system: Nasal breathing is moderately laboured; no secretions present. On auscultation, breath sounds in the lungs are harsh and evenly distributed throughout all segments; fine-bubbly rales are present in the posterobasal segments of the lungs. The cough is non-productive; no dyspnoea is noted on examination. SpO₂ 100%.

Digestive system: Palpation of the abdominal organs is painless; the abdomen is soft on palpation and accessible to deep palpation.

Spleen: not palpable.

Liver: at the edge of the costal margin.

Urinary system: urination is unobstructed; no dysuria. Urine, according to the patient, is pale yellow and clear.

Treatment administered:

Antiviral prophylaxis: Valacyclovir (500 mg tablets), 500 mg twice daily by mouth, with

24 April 2026 to 3 May 2026

Antifungal prophylaxis: Vifend 300 mg [over 1 hour] intravenously twice daily, from 24 April 2026 to 3 May 2026

Infusion therapy from 24 April 2026 to 3 May 2026

Anti-Pneumocystis prophylaxis: Co-trimoxazole (480 mg tablets), 3 tablets once daily by mouth, 24 April 2026, 27 April 2026, 29 April 2026

Adjunctive therapy: Povidone-iodine 10%, 1 treatment 6 times a day, from 24 April 2026 to 3 May 2026; Folic acid (1 mg tablets), 3 mg once daily by mouth, from 24 April 2026 to 3 May 2026;

Chlorhexidine 0.05%, once daily, 8 times per day topically, from 24 April 2026 to 3 May 2026;

Ondansetron 8 mg in up to 10 ml of saline solution, intravenously 3 times per day, from 24 April 2026 to 3 May 2026; Comforderm K (cream), 1 application 4 times daily, from 24 April 2026 to 3 May 2026; Levetiracetam (250 mg tablets), 500 mg twice daily orally, from 24 April 2026 to 3 May 2026; Vikasol 1 ml in up to 20 ml of saline solution intravenously, 24 April 2026; Bepanten (5% cream), 1 application 4 times daily, from 24 April 2026 to 3 May 2026; Allopurinol (100 mg tablets), 100 mg four times daily by mouth, from 24 April 2026 to 3 May 2026;

Antibacterial therapy: Selezolid 600 mg [over 1 hour] intravenously twice daily, from 24 April 2026 to 28 April 2026; Meronem 1.2 g in 20 ml of saline solution, intravenously three times daily, from 24 April 2026 to 3 May 2026;

Transfusion therapy: Platelet concentrate

Red blood cell suspension

Catheters: Implantable Hickman catheter – 30 March 2026 (REF: 600570)

Clinical course during hospitalisation: During the observation period, the patient's condition remained stable, with no episodes of infection. Antibiotic therapy was gradually tapered off. The transplant is not functioning; platelet concentrate and packed red blood cell transfusions were administered.

Conclusion: On 29 April 2026, a multidisciplinary consultation was held to discuss the patient's further management. Based on the results of the examination, the patient's underlying disease has progressed 14 days after CAR-T cell therapy.

Taking into account the extent of treatment administered (chemotherapy, targeted therapy, two allogeneic haematopoietic stem cell transplants and CAR-T cell therapy), the patient's poor physical condition and refractoriness to the comprehensive therapy being administered, all possible curative options within the Russian Federation have currently been exhausted. Due to the lack of alternative treatment methods, the child's condition has been deemed incurable by the multidisciplinary team. Further treatment may be exclusively symptomatic in nature, aimed at sustaining life and alleviating the patient's suffering.

The patient is being discharged from the ward for continued monitoring by palliative care specialists. Prior to discharge on 02–03 May 2026, transfusions of thromboconcentrate and packed red blood cells were administered.

Condition at the time of discharge: Clinical status: The patient is not actively complaining. The patient is feeling well. No fever. The skin is pale pink, clear and moderately moist. No signs of haemorrhagic syndrome. The mucous membranes of the oral cavity are pink, clear and moist. Cardiopulmonary function is satisfactory. No nausea or vomiting. The abdomen is soft, palpable throughout and pain-free. Bowel and bladder function are normal.

Recommendations: Recommendations:

1. Follow-up by a general practitioner/haematologist and palliative care specialists on an outpatient basis.

2. Monitoring of complete blood count with differential and blood biochemistry (urea, creatinine, potassium, sodium, ALT, AST, total and direct bilirubin, glucose, ionised calcium) – once a week; more frequently if necessary; samples to be taken at home.

3. In the event of progressive hepatosplenomegaly, leucocytosis in peripheral blood, consider prescribing mercaptopurine 50 mg/m² once daily by mouth, and administering courses of cytarabine 75–200 mg/m², vepezide 150–200 mg/m² or cyclophosphamide 150 mg/m² whilst the patient is receiving allopurinol 300 mg/day and infusion therapy comprising a 0.9% NaCl solution/5% glucose solution at a rate of 2.5–3 l/m², with the addition of 5% sodium bicarbonate (80 ml per 1,000 ml of infusion); with monitoring of urine output and blood biochemistry (potassium, sodium,

urea, creatinine, uric acid).

4. For maintenance therapy, continue treatment with FLT3 inhibitors (gilteritinib) at 40 mg once daily by mouth; the dose may be increased to 120 mg by mouth. The patient has the tablets in their possession.

5. If signs of acute PID (skin rash covering 50% or more of the body surface area, persistent vomiting, diarrhoea, or rising bilirubin levels) appear – contact the Intensive Care Unit (ICU); consider prescribing prednisolone at a dose of 1 mg/kg/day orally.

6. If pain develops, analgesia is indicated:

- If pain develops, NSAIDs are the first-line treatment. Ibuprofen may be administered at a dose of 10 mg/kg four times daily, or paracetamol at a dose of 15 mg/kg four times daily (doses for oral administration).

- If ineffective, prescribe tramadol at a dose of 2 mg/kg three times daily (intravenously); if this proves insufficient, increase the dose to 2 mg/kg four times daily (orally) in combination with NSAIDs (maximum daily dose of tramadol: 8 mg/kg/day).

- If NSAIDs and tramadol are ineffective, administer morphine subcutaneously at a starting dose of 0.1 mg/kg four times daily (strictly every 6 hours); if necessary, increase the dose to 0.1 mg/kg six times daily (strictly every 4 hours). The dose for breakthrough pain is 10–30% of the daily dose, administered as required.

- The required dose of morphine is calculated daily by adding the maintenance dose to the dose administered for breakthrough pain.

- When determining the required dose of parenteral morphine, it may be possible to switch to a transdermal patch containing fentanyl or long-acting morphine tablets.

- Switching to prolonged-release morphine tablets (MST-continus): the daily dose administered subcutaneously is multiplied by 3, and the resulting daily dose is given in two doses. Thus, if a patient is receiving 40 mg of morphine per day subcutaneously, the daily dose of MST-continus will be $40 \times 3 = 120$ mg (60 mg twice daily) orally. The tablets must not be broken!

- Switching to a transdermal system containing fentanyl: a two-step conversion.

1. The dose of subcutaneously administered morphine must be converted to an oral morphine dose (multiply the daily dose by 3). Thus, if a patient receives 40 mg of morphine per day subcutaneously, the daily dose of MST-continus will be $40 \times 3 = 120$ mg orally.

2. The resulting oral morphine dose is divided by 3 to determine the required patch dose (strength). If a patient is receiving 120 mg of morphine orally, the patch dose (strength) will be $120:3 = 40$ µg/hour. The dose is rounded up to the nearest available transdermal patch strength. In Russia, transdermal patches are available in strengths of 25 µg/hour, 50 µg/hour, 75 µg/hour and 100 µg/hour. The transdermal patch is applied for 72 hours; the patch must not be cut. During the first 18 hours after applying the first transdermal patch, routine pain relief with short-acting morphine continues.

- Analgesic doses may be gradually increased, depending on the child's needs, and are limited only by life-threatening side effects of the medicines!

- The use of promedol for chronic pain management in children is contraindicated due to the neurotoxicity of its metabolites!

7. In the event of fever, worsening pain, or the onset of signs of respiratory failure or bleeding, admission to the haematology/internal medicine department at the patient's place of residence is indicated.

8. Replacement therapy with blood products may be required – filtered (if possible, irradiated) blood products as indicated (blood group O(I)Rh(+): packed red blood cells if haemoglobin levels fall below 80 g/l, platelet concentrate when the platelet count falls below $20 \times 10^9/l$).

10. In the event of leucopenia (white blood cell count of $1.5 \times 10^9/l$) and/or neutropenia (neutrophil count of $0.5 \times 10^9/l$), provided this is not caused by treatment for the underlying condition – stimulation of granulocyte production with G-CSF at a dose of 5 µg/kg subcutaneously, monitored for haematological response!

11. Due to persistent severe secondary immunodeficiency following TGSK, should infectious episodes occur, the patient requires examination by a general practitioner or haematologist, and immediate prescription of cefixime at a dose of 400 mg/day; if oral antibiotics prove ineffective, systemic broad-spectrum antibiotics should be prescribed; due to the persistent secondary combined immunodeficiency, only intravenous injections are indicated for the patient.

12. If diarrhoea develops, add nifuroxazide 200 mg three times daily by mouth and sorbents to the treatment; if this proves ineffective, refer the patient to hospital for monitoring of fluid balance and electrolytes.

13. Where medically indicated, continue treatment as follows:

- Antiviral therapy: valacyclovir 500 mg twice daily by mouth
- Prophylaxis against Pneumocystis pneumonia: sulfamethoxazole + trimethoprim 480 mg, 4 tablets three times a week, orally
- Antifungal prophylaxis: fluconazole 300 mg daily, taken orally

Where medically indicated, consider providing the patient with these medicines immediately!!!

14. The patient has a Hickman-type central venous catheter (CVC). The patient's father has been informed of the care guidelines and possible complications! All procedures involving the CVC must be carried out under aseptic conditions! When handling the catheter, the use of syringes with a capacity of less than 10 ml is not permitted due to the risk of damaging the integrity of the catheter walls! If a central venous catheter-associated infection is suspected, the patient must be monitored in a 24-hour inpatient setting, receive immediate treatment with broad-spectrum antibiotics via a peripheral venous access, and have the central venous catheter removed! The patient must be provided with consumables for catheter management – sterile gloves, sterile wipes, catheter caps, heparin, saline solution, 10 ml syringes and antiseptics.

In contact with an infectious patient: None

Cumulative radiation dose for the current hospitalisation and previous hospitalisations (mSv): 8.2123

Sick leave certificate: No sick leave certificate is required.

Appendix No. 1 to the discharge summary: 'Examinations carried out'

Attending doctor:

Head of Department:

Chief Medical Officer:



M.A. Dunaykina

L.N. Shelikhova

Литвинов Д.В.

Please complete the questionnaire about your experience at the Centre. It will take less than 10 minutes and will help us improve our services



Point your smartphone camera at the QR code **and** follow the link

TESTS CARRIED OUT

Complete blood count

Examination	WBC		RBC		HGB		PLT		NEUT#	
Normal	4-9	6.05-9.85	4-5	4.2-4.6	115-138	130-160	180-320	204-356	1.5-8.5	2-5.5
Units	10 ⁹ /l	10 ⁹ /l	10 ¹² /l	10 ¹² /l	g/l	g/l	10 ⁹ /l	10 ⁹ /l	10 ⁹ /l	10 ⁹ /l
01.04		0.55		2.67	75			16		
02.04		0.62		3.11	87			16		
03.04		0.53		2.98	83			36		
04.04		0.58		2.79	81			28		
5 April		0.35		2.89	83			16		
06.04		0.05		2.93	81			33		
07.04		0.01		3.02	84			21		
08.04		0.01		2.81	78			14		
09.04		0.01		3.05	84			36		
10.04		0.01		3.06	86			24		
11.04		0.07		2.99	84			19		
12.04		0.05		3.01	85			38		
13.04		0.06		3.00	86			31		
14.04		0.11		2.95	82			19	0.00	
15.04		0.08		2.92	82			58		
16.04		0.10		2.85	79			42		
17.04		0.11		3.09	85			20		
18.04		0.11		2.91	83			35		
19.04		0.18		2.89	82			20		
20.04		0.17		2.80	80			33		
21.04		0.16		2.75	76			22		
22.04		0.28		3.05	85			17		
23.04		0.29		3.21	93			35	0.01	
24.04		0.41		3.06	86			25		
25.04		0.50	3.07			86		19		
26 April	0.71		2.91			82	26			
27 April	0.90		3.03			87	21			
28 April	1.23		3.03			85	40			0.06
29.04	1.68		2.92			85	28			0.15
30 April	2.16		2.90			83	19			0.23

Study	LYMPH		MONO#	EO#		BASO#
Normal range	1.2-3	1.5-7	0.09-0.6	0.002-0.3	0.061-0.5	0-0.06
Units	10 ⁹ /l	10 ⁹ /l	10 ⁹ /l	10 ⁹ /l	10 ⁹ /l	10 ⁹ /l
1 April						
02.04						
03.04						
04.04						
5 April						
6 April						
7 April						
8 April						
9 April						
10 April						
11 April						
12 April						
13 April						

Test	LYMPH		MONO#	EO#		BASO#
Normal	12-3	15-7	0.09-0.6	0.002-0.3	0.061-0.5	0-0.06
EL	10 ⁹ /l	10 ⁹ /n	10 ⁹ /n	10 ⁹ /n	10 ⁹ /l	10 ⁹ /l
14.04		0.09	0.02		0.00	0.00
15 April						
16 April						
17 April						
18 April						
19 April						
20 April						
21 April						
22 April						
23 April		0.18	0.10		0.00	0.00
24 April						
25 April						
26 April						
27 April						
28 April	0.38		0.79	0.00		0.00
29 April	0.42		1.11	0.00		0.00
30 April	0.42		1.51	0.00		0.00

White blood cell differential

Study	Reference range	Unit	22 April	27 April
Lymphocytes	19-37	%	38.00	48.00
Monocytes	3-9.9	%	12.0	51.0
Segmented neutrophils	47-72	%		1.0

Biochemical blood test

Test	Normal range	Unit	01.04	02.04	03.04	04.04	5 April	6 April	7 April
AIT	0-37	El/l		22.2				12.8	
ACT	0-39	El/l		40.2				29.6	
Albumin	32-45	g/p		37				35	
Bylshrubyn, total	0-17	μmol/l		4.2				5.3	
Total ruby pi	0-5	μmol/l		1.89				2.23	
Glucose	4.11-5.89	mmol/l		5.52				4.85	
Calcium	3.5-5.1	mmol/l	4.11	3.9	4.1	4.2	4.0	3.5	3.7
Calcium ions.	1.12-1.32	mmol/l	1.12	1.12				0.95	1
Cratsanici	0-106	μmol/l	176.1	175.7	154.5			149.3	
lyi	0-279	Pine/litre		515					
Urea	2.9-7.5	mmol/l	5.4	5.1	4.8			2.7	
Patriy	136-145	mmol/l	139.7	139	145	146	145	144	143
CRP	0-5	mg/l		6.1				7.6	
Magnesium	0.7-0.91	mmol/l		0.77				0.62	0.75
Triglycerides	0-2.3	mmol/l						1.37	
Phosphorus	0.95-1.6	mmol/l	1.0						
Cystatin C, mass concentration in serum or plasma	0.62-1.11	mg/l	1.146	1.227				0.983	
Test	Normal range	Unit	08.04	09.04	10 April	11 April	12 April	13 April	14 April
AIGT	0-37	From	9.7					121.1	
AST	0-39	Chven	26.0					109.3	
Albumin	32-45	r/n	35					38	
Total protein	60-80	g/l	56.9						
Total bilirubin	0-17	μmol/l	5.2					6.6	
Direct bilirubin	0.5	μmol/l	2.15					2.98	
1 glucose	411,589	mmol/l	4.79					5.10	

Study	Standard	Units	08.04	09 April	10 April	11 April	12 April	13 April	14 April
Calpion	3.5-5.1	mmol/l	3.8	3.4	3.8	3.9	3.7	4.2	4.4
Kalitsay non.	1.12-1.32	mmol/l	1.03	1.03				1.14	
Creatinine	0-106	μmol/l	129.1		118.9	109.3	100.0	105.9	
lyi	0-279	U/L	486						
Urea	2.9-7.5	mmol/l	4.2		3.69	4.11	4.48	4.63	
Sodium	136-145	mmol/l	142	141	139	141	137.8	137	137
CRP	0-5	mg/l	23.1						
Magnesium		mmol/l	REPLACE D	No reagent					
	0.7 - 0.91	mmol/l						0.73	
Cystatin C, mass concentration in serum or blood plasma	0.62-1.11	mg/l	0.987					0.992	

Test	Standard	Unit	15 April	16 April	17.04	20 April	21.04
ALT	0-37	Yel/n		64.4		53.2	
ACT	0-39	Units/litre		37.2		45.9	
Albumin	32-45	r/n		38		35	
Total protein	60-80	r/n		61.1			
Total bilirubin	0-17	μmol/l		5.4		6.8	
Direct bilirubin	0-5	μmol/l		2.67		3.21	
Glucose	4.11-5.89	mmol/l		5.80		5.68	
Calp	3.5-5.1	mmol/l	4.8	4.3		4.0	
Ionic potassium	1.12-1.32	mmol/l		1.17		1.12	
Creatinine	0-106	μmol/l	100.7	92.0		80.3	
lyi	0-279	U/L		324			
Urea	2.9-7.5	mmol/l	5.57	5.69		3.99	
Sodium	136-145	mmol/l	136	138		137	
CRP	0-5	mg/l	46.0	63.1		102.4	
Magnesium	0.7-0.91	mmol/l		0.74		0.74	
Ferritin	15-200	μg/l	6850.1		5941.7		7,724.5

Test	Standard	EL	22 April	23 April	25.04	27.04	29 April
ALT	0-37	U/L		106.5			
	0-41	U/L				183.7	168.0
ACT	0-39	Units/litre		135.0			
	0-40	EL/l				136.7	136.6
Albumin	32-45	r/n		38.8		37.3	36.7
Total protein	60-80	g/l		60.99		58.44	56.53
Total bilirubin	0-17	μmol/l		7.57		15.22	21.94
Total bilirubin	0-5	μmol/l		5.18		11.63	18.79
Glucose	411-5.89	mmol/l		5.67		5.48	5.59
Kalši	3.5-5.1	mmol/l	4.38	4.1	4.1	4.12	3.9
Kalitsay non.	1.12-1.32	mmol/l	1.11	1.19		1.16	1.17
Creatinine	0-106	μmol/l		69.5			62.6
Lits	0-225	U/L				536.0	573.7
	0-279	Bd/l		562.3			
Urea	2.9-7.5	mmol/l		3.30		3.32	2.81
Ipatrij	136 145	mmol/l	138.6	137	137	139	135
CRP	0-5	mg/l		61.56		33.2	31.4
Magnesium	0.7-0.91	mmol/l	0.81	0.86		0.83	0.90
Ferritin	20-2.50	μg/l				8304.2	7268.9

General urine analysis

Test	Reference range	Unit	02.04	08.04
Bacteria (sediment)	0-0		30	30

Test	Normal	EL	02 April	08.04
Protein	0-0.1	g/p	not detected	not found
Bilirubin	0-0	mmol/l	not detected	not detected
Glucose	0-0	mmol/l	within normal limits	within normal limits
Yeast (sediment)	0-0	n/a	not detected	not detected
Ketones	0-0	mmol/l	Not detected	not detected
Crystals (precipitate)		n/a		not detected
	0-0	n/a	not detected	
Round epithelium (sediment)	0-0.5	n/a	Not detected	not detected
Leukocytes	0-0	per µl	not detected	Not detected
White blood cells (sediment)	0-3	Not detected	1.80	1.80
Niprups	0-0	qualities	not detected	not detected
Relative density	1.012-1.025		1.007	1.010
Pathological casts (sediment)	0-0	n/a	not detected	not detected
Squamous epithelium (residue)	0-2	n/3	not detected	not detected
Transparency	0-0		Clear	Clear
Slime (sediment)	0-0		not detected	Not detected
Sperm (sediment)	0-0	n/a	absent	-
Urobilinogen	0-0	mmol/l	within normal range	within normal limits
Colour	0-0		Light yellow	Pale yellow
Cylinders: high clay content (residue)	0-0.1	n/3	not detected	not detected
Red blood cells	0-0	per µl	not detected	not detected
Red blood cells (sediment)	0-2	per field of view	0.80	1.00
pH	5.5-7		7	8

Biochemical urinalysis

Study	Normal	Unit	02.04
Creatinine (24-hour)	8.84 – 13.3	mmol/day	-
Urine creatinine (single sample)	3460–22900	µmol/l	2814.00

Infectious status

PCR of venous blood

Test	06.04	06.04	08.04	13 April	13 April	20 April	20 April	27 April
PCR for adenovirus		OTR			OTR		OTR	
PCR-BK virus		18	OTR					
PCR-CMV(K)	OTR			OTR		OTR		OTR
PCR-EBV(K)	OTR			OTR		OTR		OTR
PCR-HHV-VI (K)	OTR			OTR		OTR		OTR

Research	27 April
PCR-Adenovirus	Yos
PCR-BK virus	
PCR-CMV(K)	
PCR-EBV(K)	
PCR-III NV-VI (K)	

BAL fluid ISHR

Test	01.04	01.04
PCR-Chlamydomphila pneumoniae	OTR	
PCR-Mycobacterium tuberculosis complex (MTC)		OTR
PCR-Mycoplasma pneumoniae	OTR	
PCR-Pneumocystis		OTR

Test	01.04	01.04
Detection of influenza A/B virus by PCR		OTR

Microbiological (culture) test of blood for sterility (Blood Aerobes) dated 19 April 2026:

No microorganisms detected (MN)

Microbiological (culture) test of blood for sterility (Blood Aerobes) dated 20 April 2026:

No microorganisms detected (MN)

Microbiological (culture) test for blood sterility (Blood Aerobes) dated 21 April 2026:

No microorganisms detected (MN)

Microbiological (culture) test for blood sterility (Blood Aerobes) dated 22 April 2026:

No microorganisms detected (MN)

Microbiological (culture) test for blood sterility (Blood Aerobes) dated 23 April 2026:

No microorganisms detected (MN)

Microbiological (culture) test for blood sterility (Blood Aerobes) dated 27 April 2026:

No microorganisms detected (MN)

No microorganisms detected (MN)

Microbiological (culture) examination of a skin swab (Skin swab) dated 3 April 2026:

Staphylococcus haemolyticus (STFHAEM) 10*3 KOE/ml

Microbiological (culture) examination of biological material for aerobic and facultative anaerobic microorganisms (Swab/discharge from the right ear) dated 6 April 2026:

No microorganisms detected (MN)

Instrumental investigations

Ultrasound scan of the abdominal organs dated 2 April 2026

Conclusion: Ultrasound findings suggestive of diffuse changes in the renal parenchyma.

Ultrasound examination of the scrotum (with Doppler) dated 28 April 2026

Conclusion: Ultrasound findings consistent with bilateral hydrocele. Soft tissue masses in the right inguinal region.

Echocardiography dated 31 March 2026

Height, cm: 186.5 Weight, kg: 59.6 BMI, m²: 1.74 Heart rate: 82 LVEDD, mm: 57.8 Z-SCORE: 1.8 LVEDV, ml: 165.2 iLVEDV, ml: 94.9 LVEDD, mm: 40.3 EF, %: 56.9 LVESV, ml: 71.3 LVSD, mm: 8.7 Z-SCORE: 0.5 LVL, mm: 8.2 Z-SCORE: 0.6 LVIM (g/m²·7): 34.71 LVEDV, ml: 121 Z-SCORE: 0.7 LVEDV (i), ml: 69.5 LVEDV (c), ml: 52 Eo, ml: 69.0 EF, %: 57.0 Ascending aorta diameter, mm: 26 Z-SCORE: 0.4 Aortic isthmus diameter, mm: 14 Z-SCORE: -0.1 Left atrial area, cm²: 14.5 Z-SCORE: -0.1 Right ventricular area, cm²: 22 Z-SCORE: 0.7 Pulmonary artery (trunk), mm: 29 Z-SCORE: 1.4 S of the right atrium, cm²: 15 Z-SCORE: -0.2 Aortic valve Vmax, m/s: 0.74 Peak pressure, mmHg: 2.2 Peak pressure at the aortic isthmus, mmHg: 8 Mitral valve V(E), m/s: 0.7 Peak pressure, mmHg: 2.0 Pulmonary artery Vmax, m/s: 0.95 Peak pressure, mmHg: 3.6 Tricuspid valve V(E), m/s: 0.5 Peak pressure, mmHg: 1.0 Peak regurgitation pressure, mmHg: 12 Central venous pressure: 5 Right atrial pressure: 17.0 D Actual net positive pressure: 19 D max Calculated net positive pressure: 23.3

Computed tomography of the thoracic cavity (multi-slice) dated 15 April 2026

Description: Performed as a follow-up to the CT scan dated 23 March 2026. Distal end of the central venous catheter in the right atrium. On follow-up, a reduction in the volume and density of previously detected focal and infiltrative changes bilaterally is noted, with a small interstitial induration persisting. Against this background, perivascular foci of grade C1 persist on the right, measuring up to 2.5 mm and up to 5 mm, with no change in size. No other 'new' focal or infiltrative changes were detected in the lungs. The trachea and large bronchi are patent; their lumina are not deformed and their walls show no abnormalities. There is minimal content in the subsegmental bronchi. There is no fluid in the pleural or pericardial cavities. The mediastinum is situated along the midline and is not widened. The thymus is in its typical position and has a homogeneous structure. A few non-enlarged lymph nodes are visible in the mediastinum. A small number of lymph nodes ranging in size from 3 to 7 mm are visualised in the axillary regions. At the

no areas of pathological density or bone destruction were identified.

Conclusion: Incomplete resolution of bilateral inflammatory changes in the lungs. Stable C1 lesions on the right.

Helical computed tomography of the paranasal sinuses (multi-slice) dated 16 April 2026

Description: Pneumatisation of the frontal sinus is intact; the mucous membrane is not thickened. The maxillary sinuses are symmetrical and pneumatised on both sides; their mucous membrane is unevenly thickened in a circular, polypoid pattern to 10–11 mm. The main sinus is normally developed and pneumatised; the mucous membrane is unevenly thickened to 5 mm. The cells of the ethmoid bone are pneumatised, with a few cells showing a thickened mucous membrane. The nasal septum is in the midline. The mucous membrane of the nasal conchae is not thickened. The bony walls of the paranasal sinuses are visible throughout their entire length; no conclusive evidence of their destruction has been found. Right side. The mastoid process has a moderately developed cellular system, with numerous cells showing impaired pneumatisation. The maxillary sinus is of normal size; pathological contents are present within the lumen. The bony portion of the external auditory canal is correctly formed; its lumen is of normal width. The external auditory canal is unobstructed. The tympanic membrane is unremarkable. The tympanic cavity is of normal size; its pneumatisation is intact. No structural abnormalities of the ossicular chain were identified. The structures of the inner ear and the internal auditory canal are unaltered. No definitive signs of destruction of the temporal bone elements were identified. The facial nerve canal is normally situated and not dilated. The large vessels show no abnormalities. Left side. The mastoid process has a moderately developed mastoid air cell system, with numerous air cells showing impaired pneumatisation. The mastoid antrum is of normal size and pneumatised. The bony portion of the external auditory canal is correctly formed, and its lumen is of normal width. The external auditory canal is unobstructed. The tympanic membrane shows no abnormalities. The tympanic cavity is of normal size, and its pneumatisation is intact. No structural changes in the ossicular chain have been identified. The structures of the inner ear and the internal auditory canal are unaltered. No reliable signs of destruction of the temporal bone elements were identified. The facial nerve canal is normally situated and not dilated. The large blood vessels show no abnormalities.

Conclusion: Pneumatisation of the paranasal sinuses, against a background of thickened mucous membrane, is not significantly impaired. Pathological contents are present in numerous cells of the mastoid processes on both sides; pneumatisation of the tympanic cavities is not impaired bilaterally.

Computed tomography of the temporal bone (multi-slice) dated 16 April 2026

Description: Described in accordance with the standard protocol in study No. 46463095

Conclusion: Described in accordance with the standard protocol in study No. 46463095

Transcription, description and interpretation of electrocardiographic data dated 1 April 2026

Conclusion: No recording was made of lead V1. Normal position of the electrocardiographic axis. Mild sinus arrhythmia with a heart rate of 75–83 beats per minute. Borderline prolongation of the QTc interval to 0.46 seconds.

Electrocardiogram recording dated 1 April 2026

Description: examination performed by

Conclusion: see ECG interpretation

Wakefulness electroencephalogram (up to 30 minutes) dated 10 April 2026

Description: EEG recording during passive wakefulness with functional tests.

Conclusion: The EEG contains a large number of physical artefacts. The recording is invalid and cannot be interpreted.

Electroencephalography during wakefulness for up to 30 minutes, dated 15 April 2026

Description: EEG recording during passive wakefulness and the initial stages of non-REM sleep. The wakefulness EEG is organised by age. The main cortical alpha rhythm has a sufficient age-related frequency index (10 Hz) and sufficient amplitude (30–70 μ V), with a normal zonal

distribution, and is weakly modulated. In the fronto-central regions, a fragmentary sensorimotor myo-rhythm is recorded with a frequency of 10 Hz and an amplitude of up to 50 μ V. Slow waves, predominantly in the theta band and of low index, are recorded as diffuse waves, with an amplitude no higher than background levels, predominantly in the frontal regions. Beta activity of low index is recorded diffusely, with an amplitude not exceeding 25 μ V. The activation response is fairly pronounced. During the RFS, no rhythm adaptation response is observed. In response to the GV test, the baseline rhythm is rapidly restored after the test ends (within 10 seconds). Upon falling asleep, a disorganisation of the rhythm is noted, fragmentation and reduction of the main cortical rhythms, an increase in the slow-wave index and beta activity, followed by the appearance of sharp biphasic potentials predominantly in the fronto-central-parietal region ('parietal potentials'). Deeper stages of sleep are not reached. The production of physiological sleep patterns in the initial stage of non-REM sleep is adequate. **Conclusion:** The EEG during wakefulness and the initial stages of physiological sleep is within the age-appropriate normal range. There is no typical epileptiform activity in the recording. The synchronised video file shows no abnormalities.

Electroencephalogram during wakefulness for up to 30 minutes, dated 21 April 2026

Description: EEG recording during passive wakefulness. The wakefulness EEG is age-appropriate. The main cortical alpha rhythm has an age-appropriate frequency index (10–11 Hz) and sufficient amplitude (30–70 μ V), with correct zonal distribution; it is weakly modulated. In the fronto-central regions, a fragmentary sensorimotor myo-rhythm is recorded with a frequency of 10–11 Hz and an amplitude of up to 50 μ V. Slow theta-range waves of low index are recorded as diffuse waves, with an amplitude no higher than background levels, predominantly in the frontal regions. Beta activity of low index is recorded diffusely, with an amplitude not exceeding 25 μ V. The activation response is fairly pronounced. During the RFS, no rhythm adaptation response is observed. In response to the GV test, the baseline rhythm is rapidly restored after the test ends (within 10 seconds).

Conclusion: The wakefulness EEG is within the age-appropriate normal range. There is no typical epileptiform activity in the recording. The synchronised video file shows no abnormalities.

Consultations with specialists

Examination by a neurologist (S.P. Khomyakova) dated 1 April 2026

Conclusion: Secondary neuromuscular syndrome. Drug-induced neuropathy.

Examination by an otolaryngologist (O.I. Vereshchagina) on 2 April 2026

Conclusion: Acute protracted rhinosinusitis.

Acute right-sided otitis media, perforated stage.

Left-sided exudative otitis media.

Recommendations: 1) CT scan of the paranasal sinuses and temporal bones;

2) Avoid getting water in the right ear;

3) Cleaning of the right external auditory canal in an ENT clinic;

4) Compound ear drops (Dioxydin 5 ml + 0.9% NaCl 5 ml): 4–5 drops into the right external auditory canal 2–3 times a day;

5) Compound ear drops (Dioxydin 5 ml + saline solution 5 ml + Tizin 2 ml), 2–3 drops into each nostril 3–4 times a day;

6) Regular follow-up with an ENT specialist.

Examination by a nephrologist (M.E. Aksenova) on 3 April 2026

Conclusion: Acute kidney injury, stage 1–2 (30 March 2026).

eGFR (eGFR₂₅) = 55.6 ml/min/1.73 m².

Left aorto-mesenteric conflict (compression of the left renal vein).

Рекомендации: -LDG, haptoglobin, schizocytes in blood

- Monitoring of blood pressure and diuresis

- Monitoring of CRP, creatinine, cystatin C and blood electrolytes over time

- Adequate fluid intake, IV fluids (at least 1 l/m²/day + losses)

- Ongoing monitoring

Consultation with an ENT specialist (O.I. Vereshchagina) on 3 April 2026

Conclusion: Acute protracted rhinosinusitis.

Acute right-sided otitis media, chronic stage?

Right-sided otitis externa?

Left-sided exudative otitis media.

Recommendations: 1) CT scan of the paranasal sinuses and temporal bones;

2) Avoid getting water in the right ear;

Avoid getting water in the right ear;
3) Compound ear drops (Lioxydin 5 ml + 0.9% NaCl 5 ml), 4-5 drops into the right external auditory canal 2-

4) Compound ear drops (Dioxydin 5 ml + saline solution 5 ml + Tizin 2 mg), 2-3 drops into each nostril 3-4 times a day;

5) Microbiological examination of a swab from the right external auditory canal for bacterial and fungal pathogens;

6) Follow-up with an ENT specialist.

Examination by an otolaryngologist (O.I. Vereshchagina) on 09/04/2026

Conclusion: Acute protracted rhinosinusitis, clinical improvement.

Acute right-sided otitis media, reparative stage.

Left-sided exudative otitis media.

Recommendations: 1) CT scan of the paranasal sinuses and temporal bones;

2) Avoid getting water in the right ear;

2) Avoid getting water in the right ear;

3) Compound ear drops (Lioxydin 5 ml + 0.9% NaCl 5 ml), 4-5 drops into the right external auditory canal 2-

3 times a day;

4) Follow-up with an ENT specialist.

Examination by an ophthalmologist (V.A. Mishanina) dated 14 April 2026

Conclusion: preretinal haemorrhages in both eyes

Recommendations: in consultation with the attending physician – subcutaneous injections of 'Hemaza' in both eyes

routine follow-up with an ophthalmologist

Examination by an otolaryngologist (O.K. Timofeeva) dated 22 April 2026

Conclusion: Acute protracted rhinosinusitis, recovery.

Acute right-sided otitis media, recovery

Left-sided exudative otitis media, recovered.

Recommendations: Follow-up with an ENT specialist.

Examination by an ophthalmologist (V.A. Mishanina) dated 22 April 2026

Conclusion: preretinal haemorrhages in both eyes

Recommendations: in consultation with the attending physician – subconjunctival injections of 'Hemaza' in both eyes; routine follow-up with an ophthalmologist

Attending doctor: Maria Alekseyevna Dunaykina

